INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

DPT INFORMATIQUE
COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
INSA Campus LyonTech - 7 Avenue Jean Capelle
Batiment Blaise Pascal - 69621 VILLEURBANNE
Phone 0472438892
Web site : http://if.insa-lyon.fr

Operating Systems and Networking

Compiler Project

IDENTIFICATION

CODE :	IF-4-PLD-COMP
ECTS :	4.0

HOURS

Lectures :	9.0 h
Seminars :	0.0 h
Laboratory :	32.0 h
Project :	0.0 h
Teacher-student contact :	41.0 h
Personal work :	30.0 h
Total :	71.0 h

ASSESSMENT METHOD

Presentation and demonstration

TEACHING AIDS

TEACHING LANGUAGE

French

CONTACT

M. DUPONT DE DINECHIN Florent
florent.de-dinechin@insa-lyon.fr
M. GUERIN Eric
eric.guerin@insa-lyon.fr

AIMS

The purpose of this project is to write a compiler from scratch.

Target skills:
- Design, analyze and transform a formal grammar
- Use techniques and tools for lexical and syntactic analysis
- Define and exploit abstract program representations
- Produce low-level code (assembly language, code generation, application binary interface)

CONTENT

Lectures:
- Tools for lexical analysis
- Tools for syntactic analysis
- Code generation from an expression tree
- Execution environment
- Code generation for control structures
- Introduction to compiler optimization

Project:
- Design and implementation of a working compiler from a language specification
- Functional validation

BIBLIOGRAPHY

- Aho, Lam, Sethi Ullman. Compilers: Principles, Techniques and Tools.

- https://gcc.gnu.org/wiki/ListOfCompilerBooks

PRE-REQUISITE

- Notions of formal grammars, e.g. IF-4-LG
- Notions of computer architecture, e.g. IF-3-AO